THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · EASY LEVEL UNIT TEST SET 3

Chapters 1–4: **Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Which of the following pairs (x, y) is a solution of $3x + y = 11$? 1

(a) $(5, 2)$ **(b)** $(2, -5)$ **(c)** $(2, 5)$ **(d)** $(-2, 5)$

2. Which of the following is a root of $x^2 - 5x + 6 = 0$? 1

(a) 4 **(b)** -3 **(c)** 7 **(d)** 3

3. The number of prime factors (counted with repetition) of 30 is: 1

(a) 8 **(b)** 4 **(c)** 3 **(d)** 2

4. The sum of the zeroes of $x^2 - 4x - 5$ is: 1

(a) -4 **(b)** -1 **(c)** -5 **(d)** 4

5. If $x = 4, y = 2$ is a solution of $3x + y = k$, then k is: 1

(a) 10 **(b)** 24 **(c)** 14 **(d)** 18

6. In $3x^2 + 7x - 2 = 0$, the value of b is: 1

(a) 7 **(b)** -2 **(c)** 3 **(d)** -7

7. Which of the following is an irrational number? 1

(a) 36 **(b)** $\frac{6}{9}$ **(c)** $\sqrt{36}$ **(d)** $\sqrt{11}$

8. The product of the zeroes of $x^2 - 4x - 5$ is: 1

(a) -5 **(b)** 5 **(c)** 4 **(d)** -1

9. The sum of two numbers is 14 and their difference is 6. The larger number is: 1

(a) 8 **(b)** 10 **(c)** 20 **(d)** 4

10. The roots of $(x + 1)(x - 4) = 0$ are: 1

(a) 3 and -4 **(b)** -1 and 4 **(c)** 1 and -4 **(d)** -1 and -4

11. The prime factorisation of 84 is: 1

(a) $2^2 \times 3 \times 7$ **(b)** $2^3 \times 11$ **(c)** 2×43 **(d)** 2×41

12. A polynomial of degree 3 has at most how many zeroes? 1

(a) 2 **(b)** 4 **(c)** 3 **(d)** infinitely many

13. The graph of $x = 3$ is a line parallel to the: 1

(a) y-axis **(b)** line $y = x$ **(c)** x-axis **(d)** none of the axes

14. The discriminant of $2x^2 + 3x + 1 = 0$ is: 1

(a) 17 **(b)** 1 **(c)** -1 **(d)** 5

15. The decimal expansion of $\frac{7}{15}$ is: 1

(a) non-terminating repeating **(b)** terminating

(c) an integer **(d)** non-terminating non-repeating

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the value of k if $x = 2, y = 5$ is a solution of $3x + 2y = k$. 2

17. Find the discriminant of $x^2 - 6x + 8 = 0$ and state the nature of its roots. 2

18. Find the HCF of 30 and 45 by prime factorisation. 2

19. Find the value of $x^2 + 5x + 6$ at $x = -2$. 2

20. Check whether the pair $x + y = 4$, $2x + 2y = 9$ is consistent. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Solve by elimination: $2x + y = 10$ and $x - y = 2$. 3

22. Solve by factorisation: $x^2 - 4x - 5 = 0$. 3

23. Verify that HCF $\times$ LCM $=$ product of the numbers for 14 and 35. 3

24. Find a quadratic polynomial whose sum and product of zeroes are -3 and 2 respectively. 3

OR

Solve by the substitution method: $x + 3y = 12$ and $2x + y = 14$. 3

25. Find the discriminant of $x^2 - 8x + 16 = 0$ and hence the nature of its roots. 3

OR

Express $\frac{7}{25}$ as a decimal after checking its type. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Find the HCF and LCM of 196 and 280 by prime factorisation and verify that HCF $\times$ LCM $=$ product. 5

OR

Find the zeroes of $x^2 + 3x - 4$ and verify the relationship between the zeroes and the coefficients. 5

27. Solve by elimination: $3x + 2y = 14$ and $2x - 3y = 5$. 5

OR

Solve by factorisation: $3x^2 - x - 2 = 0$. 5

END OF QUESTION PAPER — Total: 50 Marks

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